Graph-Theoretic Analysis of the Human Vascular System for Bottleneck Detection and Future Blockage Prediction

ABSTRACT

The human vascular system plays a critical role in maintaining healthy blood circulation, and abnormalities such as vessel narrowing or blockage can lead to severe medical conditions. This project presents a **graph-theoretic and simulation-based approach** to analyze blood vessel networks and identify weak or high-risk vessels. The vascular system is modeled as a **directed graph**, where nodes represent vessel junctions or organs and edges represent blood vessels with attributes such as length, diameter, resistance, and flow capacity.

A **maximum flow algorithm (Edmonds–Karp)** is applied to compute blood flow from the heart to major organs and to detect flow-limiting bottlenecks. In addition, **centrality measures** are used to identify structurally important vessels within the network. A combined risk score is calculated to rank vessels based on their likelihood of blockage. Future blockage scenarios are simulated by gradually reducing vessel capacities over time. The proposed model provides an interpretable and reproducible framework for studying vascular vulnerability in a theoretical and academic context.

AIM

The aim of this project is to **analyze the human vascular system using graph theory** by modeling blood vessels as a directed network and applying **maximum flow and centrality algorithms** to identify **weak vessels, bottlenecks, and high-risk regions**, and to **simulate future blockage scenarios** under progressive capacity reduction.

CONCEPT

The human vascular system can be represented as a **directed, weighted graph**, where:

Nodes represent blood vessel junctions and organs

Edges represent blood vessels

Edge capacity represents the maximum blood flow

Edge attributes such as diameter, length, and resistance indicate vessel health

Blood circulation is treated as a **flow network problem**.
Blockage or narrowing of vessels is simulated by **reducing edge capacities over time**.
By combining **maximum flow algorithms** with **centrality analysis**, the system identifies vessels that are **structurally and functionally critical**.

METHODOLOGY

Data Collection

Vascular data is stored in CSV format with vessel properties such as length, diameter, resistance, and flow capacity.

Graph Construction

A directed graph is created where edges carry capacity and resistance attributes.

Maximum Flow Analysis

Blood flow from the heart (Aorta) to major organs is computed using a max-flow algorithm.

Bottleneck Detection

Vessels that reach full capacity during flow computation are identified as bottlenecks.

Centrality Analysis

Betweenness centrality is used to detect vessels that lie on many critical flow paths.

Risk Scoring

A combined risk score is calculated using flow utilization, vessel diameter, and centrality values.

Future Blockage Prediction

Vessel capacity is gradually reduced to simulate plaque growth and observe flow degradation.

ALGORITHM USED AND ITS CONTRIBUTION

{Edmonds–Karp algorithm}

Contribution:

Computes the **maximum possible blood flow** from the heart to organs.

Uses **Breadth-First Search (BFS)** to ensure stable and repeatable results.

Identifies **flow bottlenecks** by detecting vessels that saturate early.

Suitable for repeated simulations, making it ideal for **future blockage prediction**.

Centrality Measures (Graph Theory)

Betweenness Centrality
Identifies vessels that act as **critical connectors** in blood circulation.

Degree Centrality
Highlights vessels connected to many others, indicating high load.

Combined Contribution:
Flow algorithms identify **functional weaknesses**, while centrality measures identify **structural weaknesses**.

VISUALIZATION APPROACH

Visualization is used to clearly represent **blood flow behavior, bottlenecks, and vessel risk levels** in the vascular network.

1. Graph-Based Network Visualization

The vascular system is visualized as a **directed graph**.

Nodes represent organs or vessel junctions.

Edges represent blood vessels.

2. Color Coding of Vessels

Red edges → High-risk / bottleneck vessels

Yellow edges → Moderate risk

Green edges → Normal blood flow

Color intensity is determined using the **risk score**, which combines:

Flow utilization

Vessel diameter

Betweenness centrality

3. Flow Thickness Representation

Edge thickness is proportional to **blood flow value**.

Thicker edges indicate higher flow.

Thinner edges indicate restricted flow or narrowing.

4. Time-Based Visualization

The network is re-plotted after each simulation cycle (year-wise).

Progressive reduction in capacity visually shows **future blockage development**.

5. Supporting Charts

Line graphs show **decline in blood flow over time** for organs.

Bar charts highlight **top weak vessels** based on risk score.

CONCLUSION

This project demonstrates the effective application of **graph theory and network flow algorithms** to model and analyze the human vascular system. By representing blood vessels as a capacitated directed graph, the study successfully identifies **bottleneck vessels and structurally critical pathways**. The use of the Edmonds–Karp algorithm enables stable and repeatable maximum flow computation, while centrality analysis enhances the identification of weak vessels beyond flow values alone.

The simulation of progressive capacity reduction provides insight into **potential future blockage development**, making the model useful for predictive analysis in an academic setting. Although the study is theoretical and not intended for clinical diagnosis, it highlights the potential of graph-based methods in biomedical analysis. Future work may include the use of real medical imaging data, integration of pressure-based flow models, and validation with clinical datasets.

Paper Used for Reference

Primary Reference Paper

Reichold, J., Stampanoni, M., Keller, A. L., Buck, A., Jenny, P., & Weber, B. (2009).
Vascular graph model to simulate the cerebral blood flow in realistic vascular networks.
Journal of Cerebral Blood Flow & Metabolism, 29(8), 1429–1443.
DOI: 10.1038/jcbfm.2009.58

Comparison Between Existing Research and Our Research

"Specifically, the paper titled *Vascular graph model to simulate the cerebral blood flow in realistic vascular networks* by Johannes Reichold et al. focuses on modeling blood vessels as a graph of resistive elements and analyzing blood flow distribution under normal and occluded conditions.

Our work differs in that we explicitly use maximum flow algorithms and graph centrality measures to identify bottleneck vessels, rank vessel risk, and simulate future blockage progression. Hence, while the inspiration comes from their graph-based vascular modeling, our contribution lies in algorithmic analysis and predictive simulation."

Aspect	Their Paper (Reichold et al.)	This Paper
Max-flow algorithms	Not used	Yes (Edmonds–Karp)
Bottleneck detection	Implicit (via flow changes)	Explicit identification
Centrality measures	Not applied	Applied
Risk ranking of vessels	Not performed	Performed
Future blockage prediction	Limited occlusion analysis	Clear time-based simulation

Our work can be considered an extension of the graph-based vascular modeling approach proposed by Reichold et al. (2009), where we incorporate algorithmic flow optimization and graph centrality analysis to identify bottlenecks and predict blockage progression